

MAGARIA, NigerWMO: **610910**Lat: **12.983N**Lon: **8.933E**Elev: **403**StdP: **96.58**Time Zone: **1.00 (E01)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.3	10.9	-7.8	2.0	29.9	-6.3	2.3	28.6	7.0	23.4	5.5	23.9	0.2	40	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.7	41.9	20.8	41.0	20.8	40.0	20.7	26.3	33.3	25.9	32.9	25.6	32.5	2.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.7	29.1	24.2	20.1	28.8	23.8	19.7	28.6	84.5	33.1	82.7	33.0	81.1	32.6	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.9	4.2	6.2	44.0	1.2	0.7	5.3	44.5	4.6	44.9	3.9	45.3	3.0	45.8		
(5)			3.3	28.1	0.7	1.5	2.7	29.2	2.3	30.0	1.9	30.9	1.4	31.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	20.8	24.3	28.6	32.1	33.3	31.7	29.0	27.6	29.0	29.3	25.4	21.6
	DBStd	4.39	2.55	3.13	2.49	2.14	1.89	2.16	1.83	1.58	1.58	1.57	2.16	2.19
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	6	5	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6472	335	399	578	662	722	651	589	546	571	599	461	359
	CDD18.3	3437	82	166	319	412	464	401	331	288	321	341	211	102
	CDH23.3	45775	1228	2141	4437	6266	7131	5697	3963	2950	3689	4246	2648	1380
	CDH26.7	26713	536	1167	2818	4225	4776	3459	1942	1244	1876	2502	1546	622
Wind	WSAvg	1.5	1.5	1.5	1.5	1.5	1.8	2.1	1.8	1.3	1.1	1.1	1.1	1.3
Precipitation	PrecAvg	545	0	0	1	2	23	64	170	200	83	11	0	0
	PrecMax	879	0	14	18	28	103	177	340	404	238	74	1	0
	PrecMin	258	0	0	0	0	0	9	30	50	13	0	0	0
	PrecStd	161	0	2	3	5	26	39	69	79	49	18	0	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	39.3	41.3	43.1	43.0	41.0	37.7	35.3	38.3	39.5	38.4	36.0
		MCWB	16.6	18.6	19.3	20.2	21.3	23.4	24.0	25.6	23.3	21.1	19.2	17.7
	2%	DB	34.0	37.5	40.2	42.0	42.0	39.6	36.1	34.1	36.7	38.6	37.2	34.0
		MCWB	16.1	17.6	18.6	20.0	21.7	23.5	24.5	25.4	24.4	21.0	18.5	16.4
	5%	DB	32.1	35.8	39.1	41.1	41.1	38.5	35.0	33.2	35.5	37.6	36.0	32.2
		MCWB	15.0	17.1	18.0	19.9	21.9	23.8	24.5	25.2	24.9	21.1	18.0	15.5
	10%	DB	29.9	33.7	37.6	40.0	40.0	37.2	33.7	32.1	34.3	36.4	34.5	30.3
		MCWB	14.3	16.2	17.5	19.5	22.2	24.0	24.5	25.0	25.1	21.5	17.3	14.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	21.1	22.6	24.9	26.2	26.4	26.5	26.7	26.8	26.0	20.9	18.4
		MCDB	34.2	35.8	36.6	35.6	35.4	34.8	33.4	32.9	33.6	33.3	35.4	34.4
	2%	WB	16.6	18.6	20.6	23.9	25.3	25.7	25.9	26.1	26.2	25.2	19.3	17.1
		MCDB	32.3	34.9	35.8	35.1	35.6	34.2	32.7	32.3	32.9	33.0	34.4	32.1
	5%	WB	15.6	17.5	19.3	23.1	24.7	25.2	25.5	25.6	25.9	24.6	18.6	16.1
		MCDB	30.9	34.3	36.6	34.9	35.3	33.8	32.0	31.5	32.5	32.4	33.7	30.8
	10%	WB	14.6	16.6	18.2	22.3	24.2	24.8	25.1	25.2	25.5	24.1	17.9	15.2
		MCDB	29.0	33.0	35.7	34.5	34.8	33.4	31.4	30.7	31.9	31.5	32.7	29.1
Mean Daily Temperature Range	5% DB	MDBR	18.9	18.2	18.2	16.6	13.7	11.2	9.6	8.9	11.1	15.3	19.3	18.7
		MCDBR	21.7	20.5	20.0	17.6	15.3	12.6	10.9	10.4	12.5	16.9	20.1	21.2
		MCWBR	8.3	7.3	7.0	6.2	5.3	3.0	3.0	3.3	3.6	4.8	6.6	7.9
	5% WB	MCDBR	21.3	19.6	18.6	13.9	13.0	11.2	10.0	9.9	11.2	12.8	19.8	20.4
		MCWBR	8.5	7.2	7.0	5.2	4.5	3.3	3.4	3.8	4.0	3.8	6.8	7.8
Clear-Sky Solar Irradiance	taub	0.594	0.655	0.703	0.732	0.699	0.674	0.623	0.591	0.596	0.644	0.563	0.559	
	taud	1.657	1.522	1.445	1.435	1.510	1.602	1.757	1.825	1.797	1.636	1.699	1.720	
	Ebn at Noon	712	690	672	652	661	669	707	740	739	690	732	728	
	Edh at Noon	245	291	321	323	294	264	228	215	222	256	233	225	
All-Sky Solar Radiation	RadAvg	5.57	5.98	6.40	6.60	6.50	6.19	5.87	5.73	6.17	6.12	5.81	5.44	
	RadStd	0.23	0.17	0.29	0.22	0.26	0.22	0.19	0.19	0.16	0.20	0.18	0.25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page